Pattern Recognition Letters 173 (2023) 30–37

Contents lists available at ScienceDirect

Pattern Recognition Letters

journal homepage: www.elsevier.com/locate/patrec

Towards desirable decision boundary by Moderate-Margin Adversarial Training

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ARTICLE INFO

Editor: Maria De Marsico

Keywords:
Adversarial training
Adversarial attack
Trade-off
Decision boundary

ABSTRACT

The previous adversarial training methods tended to use a larger uniform perturbation budget to obtain an inclusive decision boundary, which improved robustness. However, this large uniform perturbation budget will bring an unnecessary increase in the margin along adversarial directions, causing heavy cross-over between natural and adversarial examples. It is not conducive to balancing the trade-off between robustness and natural accuracy. In this paper, we propose a novel adversarial training scheme, namely Moderate-Margin Adversarial Training (MMAT), to achieve a better trade-off. Specifically, we generate finer-grained adversarial examples to mitigate the cross-over between them and natural examples of neighboring classes. Meanwhile, we design a hybrid loss to learn adversarial examples and natural examples respectively to further obtain a moderate decision boundary. Extensive experiments show MMAT achieves high natural accuracy and robustness under both black-box and white-box attacks. Especially, state-of-the-art robustness and natural accuracy are achieved on SVHN.

1. Introduction

Deep neural networks (DNNs) have achieved great success in many tasks. However, recent research demonstrates that DNNs are extremely vulnerable to adversarial attacks. The so-called adversarial attack is to mislead a classifier by adding some well-designed perturbations to a natural input. Although numerous countermeasures have been proposed [1–4], adversarial training has proven to be one of the most effective methods. Nevertheless, the higher Robust Accuracy (RA) achieved by adversarial training often leads to Natural Accuracy (NA) degradation, which prompts us to exploit a better trade-off between RA and NA.

A naturally trained model can attain perfect NA, but its decision boundary is close to a large portion of the examples [5], as shown in Fig. 1(a). The small average margin leads to the vulnerability of DNNs to adversarial examples. In contrast, an adversarially trained model with a larger average margin is robust to adversarial examples. However, it will trigger a superfluous increase in the margin as compared to the desired increase required for robustness [6], as shown in Fig. 1(b). Although the adversarially trained models learned inclusive decision boundaries to correctly classify adversarial counterparts, an unwarranted increase in the margin leads to a reduction in NA [7,8]. In other words, the over-inclusiveness causes a heavy cross-over between adversarial examples and natural examples near the decision boundary, which is not conducive to balancing the RA-NA trade-off [9].

In this paper, we expect to alleviate the excessive adversarial directional margin to learn a moderate decision boundary as shown in Fig. 1(c). Thus, we propose a novel adversarial training scheme, namely Moderate-Margin Adversarial Training (MMAT). Previous work [6,10,11] mostly treats all examples equally and assigns them uniform perturbation budgets to generate adversarial counterparts. Our experiment confirmed that this is the main reason leading to excessive directional margin. We also find that the margin of each example is significantly different. So we craft finer-grained adversarial examples based on the characteristics of their margin to reduce the excessive adversarial directional margin, especially for the examples near the decision boundary. Besides, we train models with a hybrid loss function, where

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https://doi.org/10.1016/j.patrec.2023.07.009
Received 29 June 2022; Received in revised form 24 May 2023; Accepted 14 July 2023
Available online 23 July 2023
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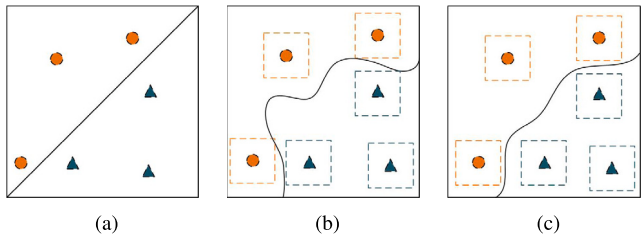


Fig. 1. A conceptual illustration of decision boundaries learned via (a) natural training, (b) standard adversarial training, and (c) moderate-margin adversarial training, respectively. The orange circle and blue triangle represent examples of different classes, and the dotted box around each example is a perturbation limit, where the adversarial counterpart is generated.

the first term is a Boosted Cross-Entropy (BCE) loss used for adversarial examples and the second term is Mean Squared Error (MSE) loss used for natural examples. Specifically, for the latter, we refer to the idea of student–teacher learning [12] to help our model adjust decision boundary. Ultimately, we obtain a moderately inclusive decision boundary, which achieves a better RA-NA trade-off. Our main contributions are summarized as follows:

(1) We reveal a reasonable margin of natural examples that plays a critical role in the RA-NA trade-off.
(2) We propose a new adversarial training method, MMAT, which reduces the heavy cross-over between natural and adversarial examples.
(3) Extensive experiments on CIFAR-10, SVHN, and ImageNet-100 show that MMAT has a higher NA while keeping a competitive RA for adversarial examples compared with other adversarial training methods.

2. Related work

Adversarial attacks. et al. [13] first observed that DNNs are highly vulnerable to adversarial examples. Goodfellow et al. [14] proposed the Fast Gradient Sign Method (FGSM) to efficiently generate adversarial examples. Madry et al. [10] proposed the Projected Gradient Descent (PGD) method as a universal "first-order adversary". Moreover, boundary-based methods like DeepFool [15] and optimization-based methods like CW [16] were also developed. Recently, the ensemble of diverse attack methods known as Auto-Attack [17], consisting of APGD-CE [17], APGD-DLR [17], FAB [18], and Square Attack [19], became a popular benchmark for testing the robustness of models. These white-box attacks assume the adversary has full knowledge of target DNNs. Numerous works have also investigated the transferability of adversarial examples for black-box attacks. Liu et al. [20] were the first to study the transferability of targeted adversarial examples. Dong et al. [21] showed that iterative attack methods incorporating the momentum term achieved better transferability. Xie et al. [22] boosted the transferability of adversarial examples by creating diverse input patterns with random resizing and random padding.

Adversarial training. As an effective defense method against adversarial attacks, plenty of research related to adversarial training has been provided. Wang et al. [23] investigated the influence of misclassified examples and proposed MART to further improve robustness. Deng et al. [24] provided the first theoretical framework to explain how adversarial training on the source data improves transfer learning. Gong et al. [25] proposed an efficient training algorithms for improving robust generalization. Jia et al. [26] proposed an example-dependent adversarial initialization to boost fast AT. Kireev et al. [27] found that AT can be successfully used to improve accuracy and calibration on common image corruptions. Yu et al. [28] proposed MLCAT prototype to avoid robust over-fitting. Besides, a large body of work has begun to focus on the trade-off between natural accuracy and robustness. Zhang et al. [11] regularized the output from natural images and adversarial examples with the KL divergence function and proposed TRADES. Zhang et al. [9] proposed searching for the least adversarial data for AT, which could be called FAT. Cui et al. [29] proposed LBGAT to adapt the logits of one model trained on natural data to guide adversarial training. Rade et al. [6] incorporated additional wrongly labeled examples during training and presented HAT to provide a notable improvement in accuracy without compromising robustness.

3. Preliminaries

Suppose $\mathbb{D} = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^N$ is a dataset where $\boldsymbol{x}_i \in \mathbb{R}^d$ is a natural example, and $y_i \in 1, 2, \ldots, K$ is its corresponding label. A DNN f_θ is a function with parameter θ to predict an input example $\boldsymbol{x}_i$: $f_\theta(\boldsymbol{x}_i) = \arg\max_{k=1,\ldots,K} \boldsymbol{z}_k(\boldsymbol{x}_i)$, where $\boldsymbol{z}_k(\boldsymbol{x}_i)$ is the logits output of the penultimate layer of the network with respect to class k. An adversarial example $\boldsymbol{x}_i' = \boldsymbol{x}_i + \boldsymbol{\delta}_i$, which misleads the DNN as $f_\theta(\boldsymbol{x}_i') \neq y_i$, while $f_\theta(\boldsymbol{x}_i) = y_i$. A small perturbation $\boldsymbol{\delta}_i$ means that $\|\boldsymbol{x}_i' - \boldsymbol{x}_i\|_p \leq \epsilon$, where $\|\cdot\|_p$ is a p-norm ($p = 1, 2$, or ∞) and ϵ is the perturbation budget.

Decision boundary. Given a classifier f_θ and a dataset $\mathbb{D} = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^N$, we define the decision boundary $\mathbb{B} \subseteq \mathbb{R}^d$ of f_θ as the set:

$$\mathbb{B} = \{\boldsymbol{x}_i \in \mathbb{R}^d | \exists l \neq j, \boldsymbol{p}_l(\boldsymbol{x}_i) = \boldsymbol{p}_j(\boldsymbol{x}_i) = \arg\max_k \boldsymbol{p}_k(\boldsymbol{x}_i)\} \tag{1}$$

Margin. Given a unit vector $\boldsymbol{v} \in \mathbb{R}^d$, the margin m at $\boldsymbol{x}_i$ along the direction $\boldsymbol{v}$ is given by:

$$m(\boldsymbol{x}_i, \boldsymbol{v}) = \arg\min_r |r| \quad s.t. \quad \boldsymbol{x}_i + r\boldsymbol{v} \in \mathbb{B} \tag{2}$$

Adversarial directions. We define the set of adversarial directions as $V = \{\boldsymbol{v}_i | \boldsymbol{v}_i = \boldsymbol{\delta}_i / \|\boldsymbol{\delta}_i\|_2\}_{i=1}^N$ where $\boldsymbol{v}_i$ is obtained by solving:

$$\boldsymbol{\delta}_i = \arg\max_{\boldsymbol{\delta}} L\left(f_\theta(\boldsymbol{x}_i + \boldsymbol{\delta}), y_i\right) \quad s.t. \quad \|\boldsymbol{\delta}\|_p \leq \epsilon \tag{3}$$

where $L(\cdot)$ is an arbitrary loss function. This optimization problem is usually solved via PGD.

4. Moderate-Margin Adversarial Training

We first describe our method at a high level here. In adversarial training, the loss function is a key component. To obtain moderate decision boundary, the loss function is divided into two parts: the first term is BCE loss used for adversarial examples, and the second term is MSE loss used for natural examples, which will be described in Section 4.1. As shown in Fig. 2, with the increase of ϵ, the examples have a more excessive directional margin, especially those near the decision boundary. Therefore, we prefer designing a finer-grained perturbation budget ϵ_i for each example to alleviate the problem and exploit the potential of larger perturbation budgets. The details are demonstrated in Section 4.2. Finally, we present our adversarial training method in Algorithm 2.

4.1. Loss function

Let $L = L_1 + L_2$ be the loss function for our adversarial training. L_1 is used for achieving a moderate margin via adversarial examples. Here we choose the BCE loss [23] instead of the commonly used CE loss:

$$L_1 = -\log\left(\boldsymbol{p}_{y_i}(\boldsymbol{x}_i')\right) - \log\left(1 - \max_{k \neq y_i} \boldsymbol{p}_k(\boldsymbol{x}_i')\right) \tag{4}$$

where $\boldsymbol{p}_k(\boldsymbol{x}_i') = e^{\boldsymbol{z}_k(\boldsymbol{x}_i')} / \sum_{j=1}^K e^{\boldsymbol{z}_j(\boldsymbol{x}_i')}$ is the probability of $\boldsymbol{x}_i'$ belonging to class k. The first term $-\log(\boldsymbol{p}_{y_i}(\boldsymbol{x}_i'))$ is the CE loss for improving robustness. The second term $-\log(1 - \max_{k \neq y_i} \boldsymbol{p}_k(\boldsymbol{x}_i'))$ is a margin term used to moderate the margin between the class with the highest confidence and the class with the second highest confidence. As is well known, these

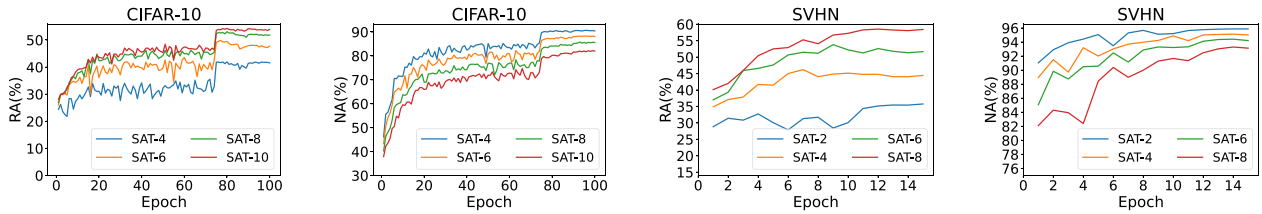
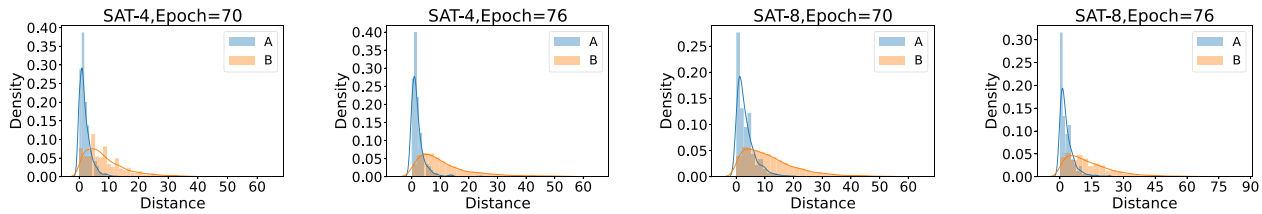


Fig. 2. The RA and NA of standard adversarially trained models. A model trained with $\epsilon = k/255$ is denoted by SAT-k. Here, we train ResNet-18 on CIFAR-10 and PreActResNet-18 on SVHN. The training details follow those of [6,30]. During training, we generate adversarial examples by using 10 steps of size $\epsilon/4$, where ϵ is the perturbation budget, and different ϵ represents different attack strengths.



(a) (b) (c) (d)

Fig. 3. The distribution of examples' distance from different decision boundaries. A represents correctly classified examples by the model, while B represents misclassified examples. Here, we pick the decision boundaries learned by SAT-4 and SAT-8 at different epochs.

two classes have the greatest similarity, which will lead to significant cross-over between natural and adversarial examples near the decision boundary.

L_2 is used for achieving a moderate margin via natural examples. Here we borrow the idea of student–teacher learning, particularly self-distillation [12] to mimic certain properties of the architecture itself. Suppose we have a well-trained *weak* robust teacher model $f_{\boldsymbol{w}}$ with high NA. The decision boundary of our target model f_θ (*i.e.*, student model) will be fine-tuned according to $f_{\boldsymbol{w}}$. This fine-tuning can pull back the excessive margin increase. Specifically, we make use of logits of $f_{\boldsymbol{w}}$ to help reshape the decision boundary of our f_θ reducing damage to NA. MSE is efficient in transferring the teacher's information to a student [31]. Therefore, we use MSE loss as L_2:

$$L_2 = \| \boldsymbol{z}(\boldsymbol{x}_i;\boldsymbol{w}) - \boldsymbol{z}(\boldsymbol{x}_i;\boldsymbol{\theta}) \|_2^2 \tag{5}$$

where $z(\boldsymbol{x}_i;\boldsymbol{w})$ and $z(\boldsymbol{x}_i;\boldsymbol{\theta})$ are the logits output of the input $\boldsymbol{x}_i$ on the models $f_{\boldsymbol{w}}$ and f_θ, respectively.

4.2. Finer-grained adversarial examples

To achieve more robustness, the previous adversarial training methods tended to use a larger perturbation budget. Meanwhile, all the adversarial examples are crafted with a uniform perturbation budget. We call these rough adversarial examples here, which will lead to the following problems in adversarial training: (1) increasing the perturbation budget is accompanied by a significant decrease in NA; (2) large perturbation budgets will cause heavy cross-over problems for misclassified examples or even correctly classified examples near the decision boundary; and (3) excessive perturbation budgets are not necessarily beneficial for robustness improvement.

Instead of using a uniform perturbation budget ϵ for all examples, we design a finer-grained budget ϵ_i for each example. Let f_{θ_t} be the model trained at epoch t. According to prediction of f_{θ_t}, natural training examples can be divided into two subsets: $\mathbb{D}_t^+ = \{\boldsymbol{x}_i | f_{\theta_t}(\boldsymbol{x}_i) = y_i\}$, and $\mathbb{D}_t^- = \{\boldsymbol{x}_i | f_{\theta_t}(\boldsymbol{x}_i) \neq y_i\}$. Fig. 3 shows a significant difference between the examples in $\mathbb{D}_t^-$ and $\mathbb{D}_t^+$, which urges us to adopt different strategies to generate adversarial counterparts for them. For the misclassified examples $\boldsymbol{x}_i \in \mathbb{D}_t^-$, they are inherently difficult to learn, so we set $\epsilon_i = 0$. For the correctly classified examples $\boldsymbol{x}_i \in \mathbb{D}_t^+$, the margin among them has its own characteristics, so we calculate the margin for each of them and then determine their specific perturbation parameters ϵ_i. Note, $\mathbb{D}_t^+$ and $\mathbb{D}_t^-$ are updated during training and ϵ_i for each example is also dynamically adjusted.

Measure of example's margin. Since calculating the exact margin in the input space is intractable, we instead search for an approximate solution. To be more precise, a small perturbation $\boldsymbol{\delta}$ is desirable such that $f_\theta(\boldsymbol{x}_i + \boldsymbol{\delta}) \neq f_\theta(\boldsymbol{x}_i)$. Ideally, we have the margin $m(\boldsymbol{x}_i, \boldsymbol{v}) \approx \|\boldsymbol{\delta}\|_\infty$, although $\boldsymbol{x}_i + \boldsymbol{\delta}$ is not necessary at the decision boundary $\mathbb{B}$. We will obtain the margin estimation with DeepFool [15]. At each iteration step of DeepFool, it linearizes the decision boundary around an example to gradually push the example over the closest boundary for minimal perturbation. This perturbation $\boldsymbol{\delta}_i$ can be written as:

$$\boldsymbol{\delta}_i = \frac{|\boldsymbol{p}_{y_j}(\boldsymbol{x}_i) - \boldsymbol{p}_{y_i}(\boldsymbol{x}_i)|}{\|\nabla \boldsymbol{p}_{y_j}(\boldsymbol{x}_i) - \nabla \boldsymbol{p}_{y_i}(\boldsymbol{x}_i)\|_1} \times \left(\nabla \boldsymbol{p}_{y_j}(\boldsymbol{x}_i) - \nabla \boldsymbol{p}_{y_i}(\boldsymbol{x}_i)\right) \tag{6}$$

with a label index:

$$y_j = \arg\min_{k \neq y_i} \frac{|\boldsymbol{p}_k(\boldsymbol{x}_i) - \boldsymbol{p}_{y_i}(\boldsymbol{x}_i)|}{\|\nabla \boldsymbol{p}_k(\boldsymbol{x}_i) - \nabla \boldsymbol{p}_{y_i}(\boldsymbol{x}_i)\|_1} \tag{7}$$

where $\boldsymbol{p}_k(\boldsymbol{x}_i)$ is the probability of $\boldsymbol{x}_i$ belonging to class k. Let I denote the stopping index of this iterative scheme for $\boldsymbol{x}_i$, *i.e.*, $f_\theta(\boldsymbol{x}_i + \sum_{i=1}^{I} \boldsymbol{\delta}_i) \neq f_\theta(\boldsymbol{x}_i)$. The desired adversarial perturbation is then defined as $\boldsymbol{\delta} = \sum_{i=1}^{I} \boldsymbol{\delta}_i$. Then we use this adversarial perturbation for the margin approximation $m(\boldsymbol{x}_i, \boldsymbol{v}) \approx \|\boldsymbol{\delta}\|_\infty$.

Crafting adversarial examples. We will use a strategy model f_s to craft adversarial examples. Since most previous adversarial training work has used SAT-8 as a baseline, here, we use a well-trained SAT-8 as a strategy model. However, different from SAT-8 using the fixed budget $\epsilon = 8/255$, we dynamically adjust ϵ according to the margin, as shown in the Algorithm 1. The margin distribution of correctly classified examples of this strategy model is shown in Fig. 3(c). We rank the margin values of the examples in ascending order and rate them into three grades according to a certain proportion. The examples are respectively divided into three sets, $\mathbb{D}_A$, $\mathbb{D}_B$, and $\mathbb{D}_C$ according to the grade. The examples in $\mathbb{D}_A$ are closest to the decision boundary (accounting for 40%), which mainly determine the improvement degree of natural accuracy. The examples in $\mathbb{D}_B$ (accounting for 30%) are the second closest to the decision boundary where the average margin is slightly greater than 8/255. The examples in $\mathbb{D}_C$ (accounting for 30%) represent those far from the decision boundary, avoiding cross-over with surrounding adversarial examples. As a result, we set three

Algorithm 1 The configuration of ϵ_i

Input: A dataset $\mathbb{D} = (\boldsymbol{x}_i, y_i)_{i=1}^N$, a strategy network f_s
Output: ϵ_i

$\mathbb{D}_A \leftarrow \varnothing$, $\mathbb{D}_B \leftarrow \varnothing$, $\mathbb{D}_C \leftarrow \varnothing$,
while $f_s(\boldsymbol{x}_i) \neq y_i$ **do**
  $\epsilon_i \leftarrow 0$
end while
while $f_s(\boldsymbol{x}_i) = y_i$ **do**
  calculating the margin of all examples: $M = \{m(\boldsymbol{x}_i)\}_{i=1}^N$
  if $m(\boldsymbol{x}_i) \leq M_{P_{40}}$ **then**
    $\mathbb{D}_A \leftarrow \mathbb{D}_A \cup \{\boldsymbol{x}_i\}$
    $\epsilon_i \leftarrow \epsilon_A = \max_m m(\boldsymbol{x}_i), \boldsymbol{x}_i \in \mathbb{D}_A$
  else if $m(\boldsymbol{x}_i) \leq M_{P_{70}}$ **then**
    $\mathbb{D}_B \leftarrow \mathbb{D}_B \cup \{\boldsymbol{x}_i\}$
    $\epsilon_i \leftarrow \epsilon_B = \frac{1}{|\mathbb{D}_B|} \sum_{\boldsymbol{x}_i \in \mathbb{D}_B} m(\boldsymbol{x}_i)$
  else if $m(\boldsymbol{x}_i) \leq M_{P_{100}}$ **then**
    $\mathbb{D}_C \leftarrow \mathbb{D}_C \cup \{\boldsymbol{x}_i\}$
    $\epsilon_i \leftarrow \epsilon_C = \min_m m(\boldsymbol{x}_i), \boldsymbol{x}_i \in \mathbb{D}_C$
  end if
end while

values of ϵ: ϵ_A, ϵ_B, and ϵ_c as shown in the Algorithm 1. Finally, the Finer-grained Adversarial Examples (FAEs) are solved by:

$$\tilde{\boldsymbol{x}}_i' = \arg\max_{\boldsymbol{x}_i'} CE\left(\boldsymbol{p}(\boldsymbol{x}_i'), y_i\right) \quad s.t. \quad \|\boldsymbol{x}_i' - \boldsymbol{x}_i\|_p \leq \epsilon \tag{8}$$

Based on the two proposed loss terms and the FAEs, we can state our final objective function for our adversarial training:

$$\min_\theta \frac{1}{N}\left(\sum_{i\in\mathbb{D}} L_1 + \sum_{i\in\mathbb{D}} L_2/\lambda\right) \tag{9}$$

Algorithm 2 Moderate-Margin Adversarial Training

Input: A dataset $\mathbb{D} = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^N$, a teacher network $f_{\boldsymbol{w}}$
Parameter: Batch-size b, learning rate η, scaling parameter λ, attack step size α and number of attack iterations T

repeat
Read a mini-batch $\{(\boldsymbol{x}_{i_j}, y_{i_j})\}_{j=1}^b$ from $\mathbb{D}$
for $j = 1, 2, \cdots, b$ **do**
  $\boldsymbol{x}_{i_j}' \leftarrow \boldsymbol{x}_{i_j} + 0.001 \times \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$, where $\mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$ is the Gaussian distribution with zero mean and identity variance
  Grade $\boldsymbol{x}_{i_j}$ based on its margin and set ϵ_{i_j} for it
  for $t = 1, 2, \cdots, T$ **do**
    $\boldsymbol{x}_{i_j}' \leftarrow \prod_{\|\boldsymbol{x}_{i_j}' - \boldsymbol{x}_{i_j}\|_p \leq \epsilon_{i_j}} \left(x_{i_j}' + \alpha \cdot sign \nabla_{\boldsymbol{x}_{i_j}'} CE)\right)$
  end for
end for
$\theta \leftarrow \theta - \frac{\eta}{b} \sum_{j=1}^b \nabla_\theta (L_1 + L_2/\lambda)$
until training completed

where λ is a tunable scaling parameter that balances the two parts of the hybrid loss. Algorithm 2 depicts our Moderate-Margin Adversarial Training (MMAT). Our strategy is based on the example margin. The teacher model is used to guide the real-time decision boundary adjustment of our model.

5. Experiment

5.1. Experimental setup

Training methods. We consider neural networks trained via Natural Training (NT), Standard Adversarial Training (SAT) [32], TRADES [11], MART [23], LBGAT [29], HAT [6], and MLCAT [28]. The trade-off parameter, β, is present in TRADES, MART, LBGAT, and HAT. A higher β indicates that more weight is given to robust accuracy.

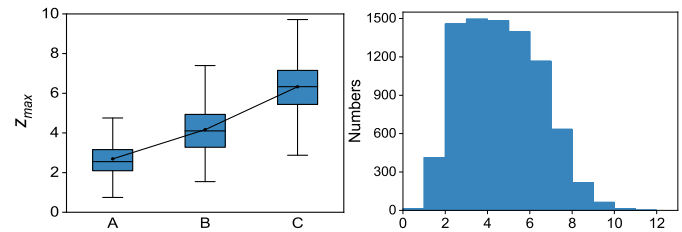


Fig. 4. The distribution of z_{max} of correctly classified examples by the strategy network. (a): The horizontal axis represents the grades based on the margin of these examples. Examples with higher grades correspond to larger z_{max}. (b): The horizontal axis represents the z_{max} value of these examples.

Table 1
The performance of ResNet-18 is trained using MMAT with different attack settings on CIFAR-10. We pick the checkpoint that has the best RA on the test set. The difference (Diff.) between best and final accuracy indicates a degradation in performance during training.

		RA			NA		
		Best	Final	Diff.	Best	Final	Diff.
Z_1	2	55.58	55.14	0.44	85.32	85.84	−0.52
	3	53.77	53.31	0.46	86.36	86.54	−0.18
	4	52.13	51.49	0.64	87.71	87.80	−0.09
ϵ_A	3/255	55.16	54.57	0.59	85.15	85.54	−0.39
	5/255	55.58	55.14	0.44	85.32	85.84	−0.52
	7/255	55.69	55.19	0.50	84.77	85.25	−0.48

Training details. For CIFAR-10, following [23], the batch size is 128 and the number of training epochs is 100. All models are trained using the SGD optimizer with Nesterov momentum 0.9 [33], weight decay 0.0035 and an initial learning rate of 0.01, which is divided by 10 at the 75th and 90th epochs. For SVHN, different from CIFAR-10, the weight decay is 0.0005 and the initial learning rate is 0.01, which is divided by 10 at the 33rd and 66th epochs. For ImageNet-100, following [27], the batch size is 128 and we run 100 epochs. We still use the SGD optimizer with momentum 0.9, weight decay 0.0005, and an initial learning rate of 0.1, which is divided by 10 at the 33rd and 66th epochs.

White-box attacks. We evaluate robustness with four types of attacks, *i.e.*, FGSM, PGD^{20}, CW$_\infty$, and Auto-Attack (AA), on three datasets. All attacks are constrained by the same perturbation budget, *i.e.*, $\epsilon = 8/255$. Note that the CW$_\infty$ attack denotes using CW-loss within the PGD framework here. For PGD, the training attack is PGD^{10} with random start and step size $\alpha = \epsilon/4$, while the test attack is PGD^{20} with random start, perturbation budget $\epsilon = 8/255$ and step size $\alpha = \epsilon/10$.

Black-box attacks. We train ResNet-18 on CIFAR-10, VGG-16 on SVHN and WideResNet-50-2 on ImageNet-100 with different adversarial training methods. Then we use the well-trained model as the source model and the target model. Adversarial examples are generated from the source model to attack the target model. PGD^{20} (black-box) is used to attack various defense models on the three datasets. And the setup follows the settings specified in the white-box attack.

5.2. Understanding the proposed MMAT

In this section, we investigate MMAT from three different perspectives: (1) the perturbation budgets of FAEs; (2) the guiding role of teacher models; and (3) replacing components of the BCE term.

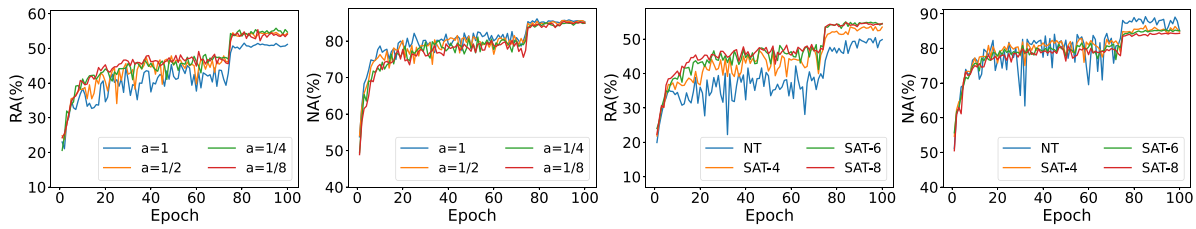


Fig. 5. Illustration of the role of teacher models. (a)–(b): Models trained with different scaling parameters a $(1/\lambda)$ under the supervision of SAT-6. (c)–(d): Models trained under the supervision of four teacher models, each representing a different level of NA and obtained through various training methods.

Table 2
Comparisons with other defense methods under white-box attacks. We pick the checkpoint that has the best RA on the test set.

Adversary	Method	CIFAR-10		SVHN		ImageNet-100	
		RA (%)	NA (%)	RA (%)	NA (%)	RA (%)	NA (%)
FGSM	SAT	57.34	83.50	60.58	89.09	40.07	67.07
	TRADES	56.72	81.57	62.21	88.49	41.18	66.80
	MART	59.68	81.39	61.36	88.27	39.83	64.36
	LBGAT	56.75	85.28	59.33	87.89	38.45	67.10
	HAT	61.10	85.95	61.21	88.70	39.94	65.27
	MLCAT	60.76	77.67	62.85	87.26	41.00	66.79
	MMAT	**59.78**	**85.32**	**62.04**	**90.65**	**40.65**	**70.87**
PGD^{20}	SAT	52.69	83.50	51.49	89.09	36.01	67.07
	TRADES	52.90	81.57	54.62	88.49	35.90	66.80
	MART	55.33	81.39	52.30	88.27	35.53	64.36
	LBGAT	52.10	85.28	51.22	87.89	34.03	67.10
	HAT	52.49	85.95	52.87	88.70	35.28	65.27
	MLCAT	56.53	77.67	52.89	87.26	36.11	66.79
	MMAT	**55.58**	**85.32**	**53.19**	**90.65**	**36.67**	**70.87**
CW_∞	SAT	49.23	83.50	45.65	89.09	29.67	67.07
	TRADES	49.41	81.57	48.56	88.49	29.84	66.80
	MART	49.91	81.39	45.09	88.27	29.43	64.36
	LBGAT	48.81	85.28	44.35	87.89	28.19	67.10
	HAT	48.73	85.95	46.17	88.70	28.96	65.27
	MLCAT	50.83	77.67	45.33	87.26	29.80	66.79
	MMAT	**50.33**	**85.32**	**47.70**	**90.65**	**30.49**	**70.87**
AA	SAT	47.67	83.50	42.59	89.09	26.57	67.07
	TRADES	48.33	81.57	46.02	88.49	27.73	66.80
	MART	48.03	81.39	41.75	88.27	26.65	64.36
	LBGAT	47.29	85.28	41.19	87.89	25.13	67.10
	HAT	47.28	85.95	43.16	88.70	25.54	65.27
	MLCAT	49.01	77.67	43.51	87.26	26.78	66.79
	MMAT	**48.49**	**85.32**	**44.61**	**90.65**	**27.19**	**70.87**

Table 3
Comparisons with other defense models under the black-box PGD^{20} attack on CIFAR-10, SVHN, and ImageNet-100. Here, MMAT is used as a source model to attack other defense models and also as a target model to be attacked by other defense models.

Dataset	Source model	Target model	RA (%)	NA (%)
CIFAR-10	MMAT	SAT	63.43	83.50
	MMAT	TRADES	63.26	81.57
	MMAT	MART	63.89	81.39
	MMAT	LBGAT	63.83	85.28
	MMAT	HAT	64.64	85.95
	MMAT	MLCAT	63.79	77.67
	SAT	MMAT	63.95	85.32
	TRADES	MMAT	63.90	85.32
	MART	MMAT	65.64	85.32
	LBGAT	MMAT	64.17	85.32
	HAT	MMAT	64.11	85.32
	MLCAT	MMAT	67.73	85.32
SVHN	MMAT	SAT	58.54	89.09
	MMAT	TRADES	60.04	88.49
	MMAT	MART	59.60	88.27
	MMAT	LBGAT	60.82	87.89
	MMAT	HAT	59.27	88.70
	MMAT	MLCAT	58.88	87.26
	SAT	MMAT	59.13	90.65
	TRADES	MMAT	62.50	90.65
	MART	MMAT	64.05	90.65
	LBGAT	MMAT	59.02	90.65
	HAT	MMAT	62.64	90.65
	MLCAT	MMAT	65.73	90.65
ImageNet-100	MMAT	SAT	57.01	67.07
	MMAT	TRADES	56.88	66.80
	MMAT	MART	55.32	64.36
	MMAT	LBGAT	56.93	67.10
	MMAT	HAT	55.30	65.27
	MMAT	MLCAT	55.25	66.79
	SAT	MMAT	57.00	70.87
	TRADES	MMAT	56.95	70.87
	MART	MMAT	60.82	70.87
	LBGAT	MMAT	57.02	70.87
	HAT	MMAT	57.54	70.87
	MLCAT	MMAT	61.49	70.87

Perturbation budgets of FAEs. Since adversarial training is time-consuming, taking 3–30 times longer to form a robust network than a non-robust equivalent [34,35], we prefer to use a metric without extra computation to determine the grade of examples instead of the margin. As shown in Fig. 4(a), z_{max} (the largest element of a network's logit output after forward-propagation) of examples meets this requirement. The distribution of z_{max} in Fig. 4(b) implies that these examples are hierarchical. Thus, we use $Z_1 = 2$ and $Z_2 = 6$ to divide these examples into three grades, *i.e.*, $\mathbb{D}_A = \{\boldsymbol{x}_i | \boldsymbol{z}_{max}(\boldsymbol{x}_i) \leq 2\}$, $\mathbb{D}_B = \{\boldsymbol{x}_i | 2 < \boldsymbol{z}_{max}(\boldsymbol{x}_i) \leq 6\}$, and $\mathbb{D}_C = \{\boldsymbol{x}_i | \boldsymbol{z}_{max}(\boldsymbol{x}_i) > 6\}$. Meanwhile, according to Algorithm 1, we have $\epsilon_A = 5/255$, $\epsilon_B = 10/255$, and $\epsilon_C = 15/255$, which means that ϵ_i of each example $\boldsymbol{x}_i$ can be set. Here we also investigate the effects of $\mathbb{D}_A$ under various Z_1 and ϵ_A, with the results summarized in Table 1. Apparently, larger Z_1 or smaller ϵ_A appears to favor NA over RA. It demonstrates the tunability of MMAT regarding the RA-NA trade-off. In addition, MMAT can enable a larger perturbation budget (15/255) to obtain models with high NA and RA, which implies that it alleviates the cross-over mixture problem [9].

Role of teacher models. We begin by looking at the parameter a $(1/\lambda)$ in our objective function defined in (9), which controls the regularization strength. A larger a indicates that the teacher model has a stronger influence on MMAT's decision boundary. As illustrated in Fig. 5(a) and (b), as a increases, the RA tends to decrease in the training process, especially when $a = 1$, whereas NA increases. For $a = 1/4$ and $a = 1/8$, the two models have higher RA and NA. Next, after fixing $a = 1/4$, we further explore the effect of different teacher models. As the teacher model becomes more robust with a lower NA, the student model also shows this trend. Thus, we confirm that the fine-tuning of a suitable teacher model facilitates the desirable course of our decision boundary to achieve a better trade-off.

Replacing components of L_1 term. Recall that L_1 is defined by BCE loss. As illustrated in Fig. 6(a) and (b), learning with CE instead of our BCE results in insufficient learning with lower robustness throughout the entire training process. Similarly, when replacing CE with KL in the inner maximization of the adversarial min–max framework, the robustness has been weak throughout the training process. Fig. 6(c) and

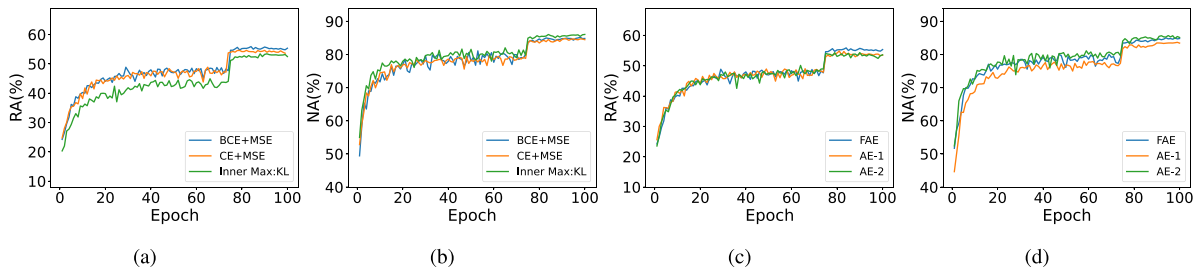


Fig. 6. Comparison of the performance of the models trained with MMAT variants. In each plot, the solid blue line represents the original MMAT. (a)–(b): We replace BCE loss with CE loss. In the adversarial min–max framework, we use KL instead of CE for inner maximization. (c)–(d): We substitute other attack strategies for our FAEs. FAE represents our original MMAT. AE-1 no longer distinguishes whether an example is correctly classified or not but instead directly ranks all examples. AE-2 adheres to the uniform perturbation budget of SAT and only attacks correctly classified examples.

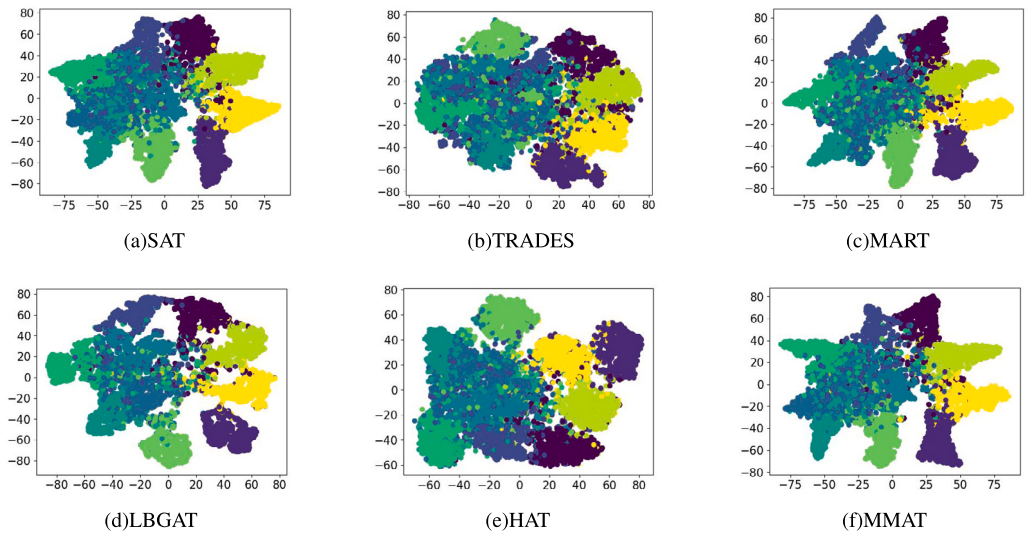


Fig. 7. T-SNE results of different methods trained on CIFAR-10. Different colors represent different classes. (a)–(f) represent different training methods, respectively.

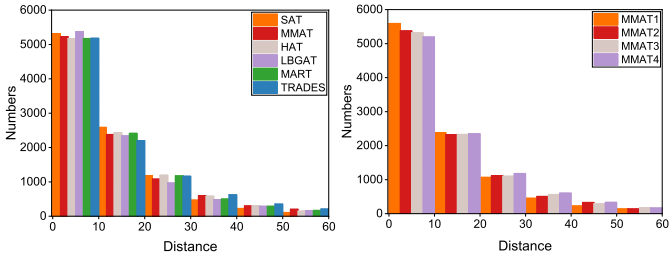


Fig. 8. The distribution of examples' margin (the values scaled to 255 times) to the decision boundaries of multiple models. (a): The models are those trained in Table 2. (b): The robustness of the four MMAT models with different λ increases sequentially.

(d) show the contribution of our BCE loss with FAEs. Specifically, we generate adversarial examples with three strategies. Both the RA and NA of AE-1 are lower than our original MMAT, suggesting that learning misclassified examples is more necessary than learning their adversarial counterparts. Moreover, AE-2 leads to a robustness degradation, indicating that our FAEs associated with characteristics of natural examples better exploit the model's robustness potential.

5.3. Comparisons with other defense methods

In this section, we assess MMAT and other defense methods mentioned in Section 5.1 against white-box and black-box attacks.

Evaluation results under white-box attacks. Table 2 shows the evaluation results of defense models under white-box attacks on different datasets. On CIFAR-10, the methods TRADES, MART, and MLCAT focus more on RA, while LBGAT and HAT tend to protect NA, and our model does both. In addition, in terms of time consumption, MLCAT takes 6 min per epoch, while all other methods are within 3 min. On SVHN, TRADES and our model are more robust under all four attacks, and our method has the best NA. Besides, the NA and RA of the remaining models are lower than MMAT. Meanwhile, TRADES and HAT take 13 min per epoch, while the other methods take at most 7.3 min (LBGAT). On ImageNet-100, the difference in RA among the models with higher RA is within 1%, while our model has the best NA. In addition, SAT is the most efficient, TRADES is the most time-consuming, and all other models take about 95 min per epoch. In general, our model achieves a better RA-NA trade-off with a relatively low time cost.

Evaluation results under black-box attacks. The evaluation results of defense models under black-box attacks on multiple datasets are reported in Table 3. Compared with the white-box (PGD^{20}) results, all defense methods achieve much better robustness against black-box attacks. Compared with other models, ours is capable of generating stronger adversarial examples. Moreover, our model exhibits high RA and NA on the three datasets, demonstrating our method's excellent performance. For a fair comparison, we keep the training and evaluation settings the same as the basic experimental setup described in Section 5.1. Hyper-parameters and training details of other defense methods are configured as per their original papers: $\beta = 6$ for TRADES and MART; ResNet18 is adopted as $M_{natural}$ for LBGAT; a well naturally

trained f_{std} for HAT. Thus, we assert that MMAT is the closest to the optimal trade-off among these classical defense methods.

6. Limitation and discussion

To further discuss the effectiveness of MMAT, we illustrate it through visualization in Fig. 7. It can be observed that our learned logit features have a larger distance between classes while being more clustered within the same class. The more distinguishable feature embedding justifies our improvement of both RA and NA. In addition, Fig. 8 exhibits the margin distributions based on multiple models. Compared with other models, our model yields a moderate margin, which is in line with our training goal. We also note that when the margin is small, the corresponding number of examples is lowest for TRADES, while as the margin increases, it has the highest number of examples. It may explain why TRADES can stand out under the strong attacks of CW_∞ and AA. Meanwhile, a limitation of our method is that generating adversarial examples is not fully dynamic. In order to guarantee actual training efficiency and effectiveness, there are artificial design steps in our generation scheme. A reasonable and efficient design deserves further investigation.

7. Conclusion

Based on previous work and our experimental findings, we discover that adversarially trained models cause an excessive increase in the margin along adversarial directions. This partly contributes to the much-debated RA-NA trade-off. The commonly used uniform perturbation budget ϵ, which ignores the characteristics of examples and leads to the severe issue of cross-over mixture, is a major contributor. To address the issue, our proposed MMAT generates finer-grained adversarial examples and designs a hybrid loss to learn adversarial examples and natural examples respectively to obtain a moderate decision boundary. Experiments show MMAT achieves a superior RA-NA trade-off compared to existing defenses.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data set used is publicly accessible.

Acknowledgments

This work was supported by Zhejiang Provincial Natural Science Foundation of China (LZ22F020007), the National Natural Science Foundation of China (92167203, 62202457), National Key R&D Program of China (2018YFB2100400), and China Postdoctoral Science Foundation, China (2022M713253).

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